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Kelas : TI-25-PA2

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<https://github.com/252310057-boop/algoritma-semster2.git>

```
1  #include <iostream>
2  #include <string>
3  #include <cstdio>
4
5  using namespace std;
6
7  #define MAX 10
8
9  class Queue {
10 private:
11     int front, rear, count;
12     string arr[MAX];
13
14 public:
15
16     Queue() {
17         front = 0;
18         rear = MAX - 1;
19         count = 0;
20     }
21
22     bool isFull() {
23         return count == MAX;
24     }
25     bool isEmpty() {
26         return count == 0;
27     }
28
29     int size() {
30         return count;
31     }
32
33     void enqueue(string x) {
34         if (isFull()) {
35             cout << ">> Maaf, Antrian penuh. Kapasitas loket sudah maksimal." << endl;
36             return;
37         }
```

```

38     rear = (rear + 1) % MAX;
39     arr[rear] = x;
40     count++;
41     cout << ">> Tiket " << x << " berhasil masuk ke dalam antrian." << endl;
42 }
43
44 string dequeue() {
45     if (isEmpty()) {
46         cout << ">> Antrian kosong. Tidak ada pengunjung yang dipanggil." << endl;
47         return "";
48     }
49     string x = arr[front];
50     front = (front + 1) % MAX;
51     count--;
52     return x;
53 }
54
55 string peek() {
56     if (isEmpty()) {
57         cout << ">> Antrian kosong." << endl;
58         return "";
59     }
60     return arr[front];
61 }
62
63 void display() {
64     if (isEmpty()) {
65         cout << ">> Antrian sedang kosong." << endl;
66         return;
67     }
68     cout << ">> Daftar Pengunjung yang Menunggu:" << endl;
69     int i = front;
70     for (int j = 0; j < count; j++) {
71         cout << " " << (j + 1) << ". " << arr[i] << endl;
72         i = (i + 1) % MAX;
73     }
74 }

```

```

75 L 1;
76
77 int main() {
78     Queue q;
79     int pilihan;
80     int nomorUrut = 1;
81
82     do {
83         cout << "\n===== " << endl;
84         cout << "        SISTEM ANTRIAN LOKET LAYANAN PUBLIK        " << endl;
85         cout << "===== " << endl;
86         cout << "1. Ambil Nomor Antrian (Enqueue)" << endl;
87         cout << "2. Panggil Pengunjung (Dequeue)" << endl;
88         cout << "3. Lihat Antrian Selanjutnya (Peek)" << endl;
89         cout << "4. Tampilkan Semua Antrian (Display)" << endl;
90         cout << "5. Cek Jumlah Antrian (Size)" << endl;
91         cout << "6. Keluar Program" << endl;
92         cout << "===== " << endl;
93         cout << "Pilih Menu (1-6): ";
94         cin >> pilihan;
95
96         char buffer[10];
97         sprintf(buffer, "%03d", nomorUrut);
98         string idTiket = buffer;
99
100        switch (pilihan) {
101            case 1:
102                if (!q.isFull()) {
103                    q.enqueue(idTiket);
104                    nomorUrut++;
105                } else {
106                    q.enqueue(idTiket);
107                }
108                break;
109            case 2:
110                {
111                    string called = q.dequeue();
112
113                    if (called != "") {
114                        cout << ">> Petugas memanggil nomor antrian: [ " << called << " ]. Silakan menuju loket." << endl;
115                    }
116                    break;
117            case 3:
118                {
119                    string next = q.peek();
120                    if (next != "") {
121                        cout << ">> Nomor antrian yang akan dipanggil selanjutnya: [ " << next << " ]" << endl;
122                    }
123                    break;
124            case 4:
125                q.display();
126                break;
127            case 5:
128                cout << ">> Jumlah pengunjung dalam antrian saat ini: " << q.size() << " orang." << endl;
129                if (q.isFull()) cout << ">> Status: Antrian PENUH!" << endl;
130                else if (q.isEmpty()) cout << ">> Status: Antrian KOSONG!" << endl;
131                break;
132            case 6:
133                cout << ">> Terima kasih." << endl;
134                break;
135            default:
136                cout << ">> Pilihan tidak Tersedia." << endl;
137                break;
138        }
139    } while (pilihan != 6);
140
141    return 0;
142 }

```

```
=====
      SISTEM ANTRIAN LOKET LAYANAN PUBLIK
=====
1. Ambil Nomor Antrian (Enqueue)
2. Panggil Pengunjung (Dequeue)
3. Lihat Antrian Selanjutnya (Peek)
4. Tampilkan Semua Antrian (Display)
5. Cek Jumlah Antrian (Size)
6. Keluar Program
=====
Pilih Menu (1-6): 3
>> Nomor antrian yang akan dipanggil selanjutnya: [ A001 ]

=====
      SISTEM ANTRIAN LOKET LAYANAN PUBLIK
=====
1. Ambil Nomor Antrian (Enqueue)
2. Panggil Pengunjung (Dequeue)
3. Lihat Antrian Selanjutnya (Peek)
4. Tampilkan Semua Antrian (Display)
5. Cek Jumlah Antrian (Size)
6. Keluar Program
=====
Pilih Menu (1-6): 6
>> Terima kasih.

-----
Process exited after 16.34 seconds with return value 0
Press any key to continue . . .
```